An explicit quartic counterexample to Zhao's Vanishing Conjecture

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ABSTRACT

Starting from the newly announced three-dimensional counterexample to the Jacobian conjecture, we explicitly carry out the Bass-Connell-Wright degree reduction and the de Bondt-van den Essen symmetric reduction. Factor reuse gives a cubic homogeneous Keller map in 27 variables. Symmetrization gives a homogeneous quartic Hessian-nilpotent polynomial in 54 variables, with 598 monomials, whose gradient Keller map is noninjective. Therefore Zhao's Vanishing Conjecture fails for this specific polynomial. The expanded polynomial and a two-point collision are supplied as exact certificates.

1. Main result

Let h: C^{27} -> C^{27} be the cubic homogeneous map constructed below. For A,B in C^{27}, set

$$\mathbf{P}(A,B) = i \,\mathrm{sum}_{i=1}^{27} h_i(A+iB)B_i. \quad (1)$$

> **Theorem 1.** The polynomial **P** is homogeneous of degree four, has 598 monomials over Q(i), and has nilpotent Hessian. The map Gamma(Z)=Z-grad **P**(Z) has Jacobian determinant one and is noninjective. Consequently $\mathrm{Delta}^m\mathbf{P}^{m+1}$ is nonzero for infinitely many m.

The construction, ordinary differentiation, and the listed collision form a finite exact certificate. The accompanying JSON file is the fully expanded polynomial, not merely a symbolic reference to the construction.

2. Normalization and stable degree reduction

Put u=1+xy. Postcomposing the announced map by the inverse of its linear part gives the identity-linear Keller map Phi:

$$\mathrm{Phi}_1=x-(3/2)x^2y-(1/2)x^3z,$$
$$\mathrm{Phi}_2=y+3xu^2z+3xy^2(4+3xy),$$
$$\mathrm{Phi}_3=u^3z+y^2u(4+3xy). \quad (2)$$

It has determinant one and sends (0,0,-1/4), (1,-3/2,13/2), and (-1,3/2,13/2) to (0,0,-1/4). To cancel a term cPQ in coordinate k, add variables a,b and replace

$$M_k \text{ -> } M_k-c(a+P)(b+Q), \quad (a,b) \text{ -> } (a+P,b+Q). \quad (3)$$

This is a pre- and post-composition by triangular automorphisms. Existing output factors may be reused with a one-variable version of (3). Apply the following six operations in order:

step	target	new or reused factors	c
1	Phi1	P=x^2; Q=xz+3y; add a1,b1	-1/2
2	Phi2	P=3x^2y; Q=2z+xyz+3y^2; add a2,b2	1
3	Phi2	P=xy; Q=a2z+3xb2; add a3,b3	-1
4	Phi3	P=xy^2; Q=7y+3xz+3xy^2+x^2yz; add a4,b4	1
5	Phi3	add a5+a4xy; reuse output b1+xz+3y	-1
6	Phi3	reuse output a3+xy; add b6+a4b1-yb4	1

Finally shear the b_4 output by M_{b4} -> M_{b4}-$M_{a3}M_{b1}$. The result has 13 variables X=($x,y,z,a_1,b_1,\ldots,a_5,b_6$) and degree three.

3. The 13-variable collision

The linear part L is the identity except that the b_1, b_2, and b_4 outputs have additional linear terms $3y$, $2z$, and $7y$. Thus det L=1. Write

$$\mathrm{Psi}=L^{-1}M=X+H_2(X)+H_3(X). \quad (4)$$

Then det J Psi=1. The lifted collision is

$$p_0=(0,0,-1/4,0,0,0,1/2,0,0,0,0,0,0),$$
$$p_1=(1,-3/2,13/2,-1,-2,9/2,-10,3/2,3/4,-9/4,-6,-27/8,9/2),$$
$$\mathrm{Psi}(p_0)=\mathrm{Psi}(p_1)=(0,0,-1/4,0,0,0,1/2,0,0,0,0,0,0). \quad (5)$$

Equations (2)-(4) are a straight-line specification of every component of Psi; the verification script expands them and checks (5) exactly.

4. A cubic homogeneous map in 27 variables

For X,Y in C^{13} and t in C, define

$$h(X,Y,t)=(tH_2(X)+t^2Y, -H_3(X), 0). \quad (6)$$

All components are cubic homogeneous. If $B(W)=W+h(W)$ and $r_j=(p_j,H_2(p_j),1)$, then $B(r_0)=B(r_1)=(p_0,0,1)$.

> **Lemma 2.** The polynomial matrix Jh is nilpotent.
> *Proof*. Homogeneity and det J Psi=1 give $\det(I+t\mathrm{J}H_2+t^2\mathrm{J}H_3)$=det J Psi($tX$)=1. The block determinant of $W \to W+h(W)$ is the same expression. Hence $\det(I+\mathrm{J}h)=1$. Since $\mathrm{J}h(sW)=s^2\mathrm{J}h(W)$, its characteristic polynomial is a pure power, proving nilpotence.

5. Symmetrization and a finite collision certificate

De Bondt and van den Essen associate $f_h(A,B)=-i \text{ sum } h_j(A+iB)B_j$ to any map h. Their characteristic-polynomial identity says Hess f_h is nilpotent exactly when Jh is nilpotent. Since $\mathbf{P}=-f_h$, Lemma 2 proves the Hessian claim.

Let $S(x,y)=(x-iy,y)$. Direct differentiation gives

$$S^{-1} \text{ Gamma } S(x,y) = (x+h(x), (I+\mathrm{J}h(x)^{\mathrm{T}})y-ih(x)). \quad (7)$$

For $K_j=I+\mathrm{J}h(r_j)^{\mathrm{T}}$, take $y_1=0$ and $y_0=K_0^{-1}(-ih(r_1)+ih(r_0))$. Then the two distinct points $S(r_j,y_j)$ collide under Gamma. Their 54 coordinates, of height at most 261, are in collision.json. The verifier differentiates the 598-term expansion and checks this equality over Q(i), with no floating-point arithmetic.

6. Failure of the Vanishing Conjecture

Zhao's inversion formula for a Hessian-nilpotent polynomial P writes the inverse of Z-t grad P as Z+t grad Q_t, where

$$Q_t = \text{sum}_{m>=0} [t^m / (2^m m!(m+1)!)] \text{ Delta}^m P^{m+1}. \quad (8)$$

If the terms in (8) vanished eventually for $\mathbf{P}$, then Q_t would be polynomial. At t=1 this would give a polynomial inverse of Gamma, contradicting the explicit collision. Thus $\text{Delta}^m\mathbf{P}^{m+1}$ is nonzero infinitely often.

7. Scope and references

Zhang's consequence note of 20 July 2026 observes existentially that the new three-dimensional counterexample makes the Vanishing Conjecture false in some dimension. The contribution here is the explicit quartic, its finite collision certificate, and a factor-reusing reduction that holds the dimension to 54. Priority and the absence of a smaller simultaneous construction must be rechecked immediately before posting.

[1] H. Bass, E. H. Connell, D. Wright, Bull. Amer. Math. Soc. 7 (1982), 287-330. DOI 10.1090/S0273-0979-1982-15032-7.

[2] M. de Bondt, A. van den Essen, Proc. Amer. Math. Soc. 133 (2005), 2201-2205. DOI 10.1090/S0002-9939-05-07570-2.

[3] W. Zhao, Trans. Amer. Math. Soc. 359 (2007), 249-274. arXiv:math/0409534.

[4] Z. Zhang, Direct Consequences of the Three-Dimensional Counterexample to the Jacobian Conjecture, 20 July 2026.

Disclosure: Alec Kriebel is a complete amateur and cannot independently verify the mathematical claims in this note. The construction, proof organization, verifiers, and typeset draft were developed with heavy assistance from ChatGPT 5.6 Sol. Independent expert review is required.